Abstract

This thesis presents the development and characterization of the experimental system required for the preparation, optical trapping, and controlled transport of cold ^{87}Rb atoms toward a Hollow-Core Photonic Crystal Fiber (HCPCF).
First, the atomic sample preparation was optimized: the optical molasses yielded a minimum temperature of $T = (8.4 \pm 0.2)\,\mu$K. Subsequent loading into an Optical Dipole Trap (ODT), improved by translating the cloud position, achieved a loading efficiency of $(0.161 \pm 0.005)\,\%$ and a trapped population of approximately $(1.0 \pm 0.1) \times 10^5$ atoms.
To implement the optical conveyor belt, a second counter-propagating 1064 nm field was established using a polarization-maintaining single-mode fiber. Characterization of the transmitted field confirmed high power and polarization stability (with fractional power and angular Allan deviations reaching 8×10^{-4} and 7×10^{-5} rad, respectively), while also identifying low-frequency technical noise and slow environmental drifts.
To counteract potential heating during optical transport, Electromagnetically Induced Transparency (EIT) cooling was investigated numerically via a quantum master-equation model in QuTiP. Multilevel simulations of the ^{87}Rb D_2 transition demonstrated that EIT cooling can successfully bring the atoms near the motional ground state, provided an active repump field is included to prevent optical pumping into uncoupled states.
These results establish the main experimental and numerical prerequisites for a future implementation of conveyor-belt transport and sub-Doppler cooling of cold atoms inside the HCPCF.